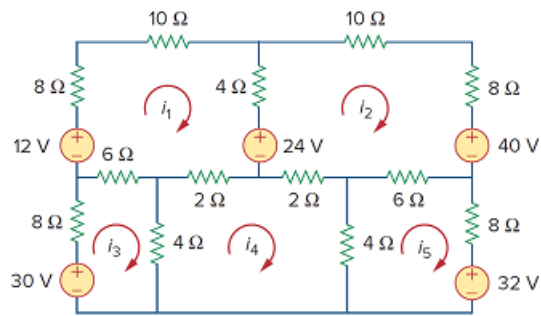


Problem

Write a set of mesh equations for the circuit in Fig. 3.110. Use *MATLAB* to determine the mesh currents.

Figure 3.110:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

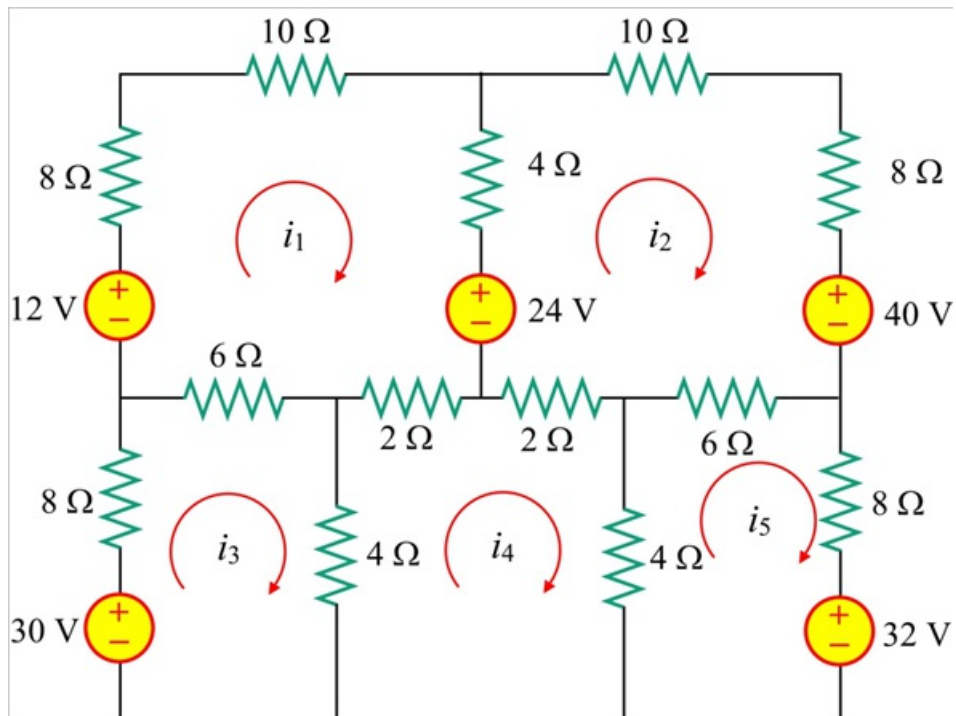


Figure 1

Step 2 of 5

Apply Kirchhoff's voltage law to the loop 1.

$$(8+10)i_1 + 4(i_1 - i_2) + 2(i_1 - i_4) + 6(i_1 - i_3) = 12 - 24$$

$$(18+4+2+6)i_1 - 4i_2 - 6i_3 - 2i_4 = -12$$

$$30i_1 - 4i_2 - 6i_3 - 2i_4 = -12$$

$$15i_1 - 2i_2 - 3i_3 - i_4 = -6 \dots\dots (1)$$

Apply Kirchhoff's voltage law to the loop 2.

$$4(i_2 - i_1) + (8+10)i_2 + 6(i_2 - i_5) + 2(i_2 - i_4) = 24 - 40$$

$$-4i_1 + (4+18+6+2)i_2 - 2i_4 - 6i_5 = -16$$

$$-4i_1 + 30i_2 - 2i_4 - 6i_5 = -16$$

$$2i_1 - 15i_2 + i_4 + 3i_5 = 8 \dots\dots (2)$$

Step 3 of 5

Apply Kirchhoff's voltage law to the loop 3.

$$8i_3 + 6(i_3 - i_1) + 4(i_3 - i_4) = 30$$

$$-6i_1 + (8+6+4)i_3 - 4i_4 = 30$$

$$-6i_1 + 18i_3 - 4i_4 = 30$$

$$3i_1 - 9i_3 + 2i_4 = -15 \dots\dots (3)$$

Apply Kirchhoff's voltage law to the loop 4.

$$4(i_4 - i_3) + 2(i_4 - i_1) + 2(i_4 - i_2) + 4(i_4 - i_5) = 0$$

$$-2i_1 - 2i_2 - 4i_3 + (4+2+2+4)i_4 - 4i_5 = 0$$

$$-2i_1 - 2i_2 - 4i_3 + 12i_4 - 4i_5 = 0$$

$$i_1 + i_2 + 2i_3 - 6i_4 + 2i_5 = 0 \dots\dots (4)$$

Step 4 of 5

Apply Kirchhoff's voltage law to the loop 5.

$$4(i_5 - i_4) + 6(i_5 - i_2) + 8i_5 = -32$$

$$-6i_2 - 4i_4 + (4+6+8)i_5 = -32$$

$$-6i_2 - 4i_4 + 18i_5 = -32$$

$$3i_2 + 2i_4 - 9i_5 = 16 \dots\dots (5)$$

Write equations (1), (2), (3), (4), and (5) in the matrix form:

$$\begin{bmatrix} 15 & -2 & -3 & -1 & 0 \\ 2 & -15 & 0 & 1 & 3 \\ 3 & 0 & -9 & 2 & 0 \\ 1 & 1 & 2 & -6 & 2 \\ 0 & 3 & 0 & 2 & -9 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} -6 \\ 8 \\ -15 \\ 0 \\ 16 \end{bmatrix}$$

Step 5 of 5

The following is the MATLAB code for calculating loop currents.

```
>> a=[15 -2 -3 -1 0;2 -15 0 1 3;3 0 -9 2 0;1 1 2 -6 2;0 3 0 2 -9];
```

```
b=[-6;8;-15;0;16];
```

```
c=inv(a)*b
```

MATLAB output:

```
c =
```

```
-0.2779
```

```
-1.0488
```

```
1.4682
```

```
-0.4761
```

```
-2.2332
```

The values of the mesh currents are,

$$i_1 = -277.9 \text{ mA}$$

$$i_2 = -1.0488 \text{ A}$$

$$i_3 = 1.4681 \text{ A}$$

$$i_4 = -476.1 \text{ mA}$$

$$i_5 = -2.2332 \text{ A}$$

Therefore, the values of the mesh currents, i_1, i_2, i_3, i_4, i_5 in the circuit shown in Figure 1 are

$-277.9 \text{ mA}, -1.0488 \text{ A}, 1.4681 \text{ A}, -476.1 \text{ mA}, -2.2332 \text{ A}$
--

 respectively.